

Separable Equations

ANSWER KEY



Order and verifying solutions

- 1 State the order of the differential equation $y'' + 3y' - 2y = \sin x$.
Second order (the highest derivative appearing is y'').
- 2 Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.
 $y' = 2e^{2x}$, so $y' - 2y = 2e^{2x} - 2e^{2x} = 0$. Yes, it is a solution.

Recognizing and solving separable equations

- 3 Solve $\frac{dy}{dx} = xy$.
Separate: $\frac{dy}{y} = x dx$. Integrate: $\ln |y| = \frac{x^2}{2} + C$. So $y = Ae^{x^2/2}$.
- 4 Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.
Separate: $\frac{dy}{y^2} = \frac{dx}{x}$. Integrate: $-\frac{1}{y} = \ln x + C$. So $y = \frac{-1}{\ln x + C}$.

Initial value problems

- 5 Solve $\frac{dy}{dx} = 3x^2y$, $y(0) = 2$.
 $\frac{dy}{y} = 3x^2 dx \Rightarrow \ln |y| = x^3 + C \Rightarrow y = Ae^{x^3}$. Since $y(0) = 2$, $A = 2$: $y = 2e^{x^3}$.
- 6 Solve $\frac{dy}{dx} = -2xy$, $y(0) = 5$.
 $\frac{dy}{y} = -2x dx \Rightarrow \ln |y| = -x^2 + C \Rightarrow y = Ae^{-x^2}$. Since $y(0) = 5$, $A = 5$: $y = 5e^{-x^2}$.

Application: radioactive decay and population growth

- 7 A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k .
 $A = 100e^{-kt}$. Then $50 = 100e^{-5k} \Rightarrow e^{-5k} = \frac{1}{2} \Rightarrow -5k = \ln \frac{1}{2} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$ per unit time.

- 8 A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.

$$P(t) = 200e^{0.03t}, \text{ so } P(10) = 200e^{0.3} \approx 269.97.$$

Further practice: separable equations

- 9 Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.

$$\text{Separate: } \frac{dy}{1+y^2} = \frac{dx}{1+x^2}. \text{ Integrate: } \arctan y = \arctan x + C.$$

- 10 Solve $y' = y \cos x$, $y(0) = 1$.

$$\frac{dy}{y} = \cos x \, dx \Rightarrow \ln |y| = \sin x + C \Rightarrow y = Ae^{\sin x}. \text{ Since } y(0) = 1, A = 1: y = e^{\sin x}.$$

More separable equations

- 11 Solve $\frac{dy}{dx} = \frac{x^2}{y}$, $y > 0$.

$$y \, dy = x^2 \, dx \Rightarrow \frac{y^2}{2} = \frac{x^3}{3} + C \Rightarrow y^2 = \frac{2}{3}x^3 + C'.$$

- 12 Solve $\frac{dy}{dx} = e^{x-y}$.

$$\text{Rewrite as } e^y \, dy = e^x \, dx. \text{ Integrate: } e^y = e^x + C, \text{ so } y = \ln(e^x + C).$$

A separable equation solved via partial fractions

- 13 Solve $\frac{dy}{dx} = y(1-y)$ (the logistic equation) using $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

$$\int \left(\frac{1}{y} + \frac{1}{1-y} \right) dy = \int \frac{1}{x} dx \Rightarrow \ln |y| - \ln |1-y| = x + C \Rightarrow \ln \left| \frac{y}{1-y} \right| = x + C. \text{ So } \frac{y}{1-y} = Ae^x, \text{ giving}$$
$$y = \frac{Ae^x}{1 + Ae^x}.$$

- 14 Solve $\frac{dy}{dx} = 2xy^2$, $y(0) = 1$.

$$\frac{dy}{y^2} = 2x \, dx \Rightarrow -\frac{1}{y} = x^2 + C \Rightarrow y = \frac{-1}{x^2 + C}. \text{ Since } y(0) = 1, C = -1: y = \frac{1}{1-x^2}.$$

Slope fields and qualitative behavior

- 15 For $\frac{dy}{dx} = x - y$, what does the slope field look like along the line $y = x$?

Along $y = x$: $\frac{dy}{dx} = x - x = 0$, so every slope-field arrow on that line is horizontal.

- 16 For $\frac{dy}{dx} = y^2$, explain why any solution with $y(0) > 0$ must be increasing for as long as it's defined.

$y^2 \geq 0$ always, and $y^2 > 0$ whenever $y \neq 0$, so $\frac{dy}{dx} \geq 0$ everywhere the solution is defined — a solution that starts positive never decreases.

Verifying a solution to an applied model

- 17 Newton's Law of Cooling states $\frac{dT}{dt} = -k(T - T_{\text{room}})$. Verify that $T(t) = T_{\text{room}} + (T_0 - T_{\text{room}})e^{-kt}$ solves it.

$T'(t) = -k(T_0 - T_{\text{room}})e^{-kt}$. Since $T(t) - T_{\text{room}} = (T_0 - T_{\text{room}})e^{-kt}$, this equals $-k(T(t) - T_{\text{room}})$, matching the equation.

A final applied separable model

- 18 A cup of coffee at 90°C is placed in a 20°C room. Newton's Law of Cooling gives $T(t) = 20 + 70e^{-kt}$. If the coffee cools to 70°C after 10 minutes, find k .

$$70 = 20 + 70e^{-10k} \Rightarrow 50 = 70e^{-10k} \Rightarrow e^{-10k} = \frac{5}{7} \Rightarrow k = \frac{\ln(7/5)}{10} \approx 0.0336 \text{ per minute.}$$

One more round of separable-equation practice

- 19 Solve $\frac{dy}{dx} = \frac{x+1}{y}$, $y(1) = 2$.

$$y \, dy = (x+1) \, dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + x + C \Rightarrow y^2 = x^2 + 2x + 2C. \text{ Using } y(1) = 2: 4 = 1 + 2 + 2C \Rightarrow 2C = 1, \text{ so}$$

$$y^2 = x^2 + 2x + 1 = (x+1)^2, \text{ giving } y = x+1 \text{ (taking the branch matching } y(1) = 2).$$

- 20 Solve $\frac{dy}{dx} = \frac{y}{x+1}$, $x > -1$.

$$\frac{dy}{y} = \frac{dx}{x+1} \Rightarrow \ln|y| = \ln|x+1| + C \Rightarrow y = A(x+1).$$

A final pair of separable equations

21 Solve $\frac{dy}{dx} = \frac{y}{x}, x > 0$.

$$\frac{dy}{y} = \frac{dx}{x} \Rightarrow \ln |y| = \ln x + C \Rightarrow y = Ax.$$

22 Solve $\frac{dy}{dx} = 3y, y(0) = 4$.

$$\frac{dy}{y} = 3 dx \Rightarrow \ln |y| = 3x + C \Rightarrow y = Ae^{3x}. \text{ Since } y(0) = 4, A = 4: y = 4e^{3x}.$$